RRGE-RAG:Radiology Report Generation Enhancement Using Hybrid Retrieval Augmented Generation

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Abstract

Medical imaging is widely used in clinical diagnosis and treatment, but writing radiology reports is time-consuming and requires substantial expertise. Automatically generating such reports is therefore important for reducing radiologists' workload and advancing clinical artificial intelligence. In this paper, we propose a **Hybrid Retrieval-Augmented Generation known as Hybrid RAG** framework for radiology report generation on the **Indiana University(IU) Chest X-Ray** dataset. Unlike prior methods that rely on a single retrieval source, our approach combines visually similar X-ray images, semantically related reports, and structured clinical knowledge to provide richer and more reliable context for generation. The retrieved evidence is fused and incorporated into a multimodal generator built on a vision encoder and a language model. By grounding report generation in hybrid retrieved evidence, the proposed method aims to improve clinical accuracy, enhance factual consistency, and reduce hallucinations. Experimental results on IU X-Ray are expected to show that our approach outperforms unimodal retrieval and non-retrieval baselines in generating coherent and clinically relevant reports.

1 Introduction

A major medical application of natural language generation (NLG) is the development of assistive systems that analyze patient X-ray images and automatically produce a textual report describing the clinical findings visible in the images. (Jing et al., 2018)However, medical report writing requires specialized domain expertise, and only experienced radiologists can accurately interpret chest X-ray images and describe the corresponding findings in a coherent manner. [1]An automatic chest X-ray report generation system can reduce the workload of radiologists and are in urgent need (Brady et al., 2012). Recent multimodal foundation models have exhibited remarkable capabilities in challenging healthcare tasks, motivating an automation of this process to enhance physicians' efficiency on clinical decision-making and improve patient health outcomes. (Sun et al., 2025)
newline Although prior medical multimodal foundation models have demonstrated promising capabilities on report generation given the radiology image, they still suffer from serious hallucinations by generating factually inaccurate reports (Pal et al., 2023; Pal and Sankarasubbu, 2024; Ahmad et al., 2023). Factual correctness is especially critical in chest radiology domains, as minute textual differences can drastically invert radiology report meaning and downstream prescribed treatments (Delbrouck et al., 2022; Song et al., 2024; Liu et al., 2024). Retrieval-Augmented Generation (RAG) has emerged as a popular paradigm to address this issue by grounding text generation with retrieved relevant knowledge given a query (Lewis et al., 2020; Wolf et al., 2024; Gao et al., 2023).However, developing medical multimodal retrievers remains challenging, requiring retrievers to bridge the gap between symptomatic image semantics and factually-equivalent report text.

To capture fine-grained details in chest radiographs and improve the factual completeness of generated reports we introduce **RRGE-RAG** it is framework takes as input a patient's chest X-ray images, including both frontal and lateral views, together with structured clinical labels derived from the Problems column. Given these inputs, the system performs retrieval to identify the Top-K most similar prior patient cases based on image similarity. Each retrieved case provides complementary contextual information, including the corresponding frontal and lateral chest X-ray

[1]https://github.com/Vibolrottanaseng/NLU-Project-Wave2Vec

images, the associated radiology report composed of Findings and Impression sections, and the structured clinical labels. By leveraging these retrieved examples as external evidence, the model is guided to generate a clinically grounded report for the input case. The primary generation target is the Impression section, since it summarizes the key diagnostic conclusion in a concise and clinically useful form, while generation of the Findings section can be considered as an optional extension. This design allows the framework to combine visual evidence from the current patient with relevant knowledge from similar historical cases, with the goal of improving factual accuracy, reducing hallucination, and producing more reliable radiology reports as shown in Figure 1.

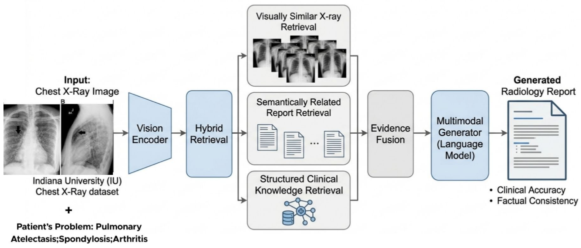


Figure 1: An Overview of RRGE-RAG framework. Hybrid RAG framework for chest X-ray report generation. A chest X-ray from the Indiana University dataset is passed through a vision encoder, then a hybrid retrieval module retrieves three types of supporting evidence: visually similar X-rays, semantically related reports, and structured clinical knowledge. These retrieved evidences are combined through evidence fusion and sent to a multimodal language generator, which produces the final radiology report with improved clinical accuracy and factual consistency.

We have came up with 3 research question for this study:

1. How does the integration of retrieval-augmented generation (RAG) influence the accuracy and clinical relevance of automated radiology report generation compared to traditional image-to-text models?

2. To what extent does multimodal fusion of X-ray images, clinical labels, and retrieved case information improve the coherence and completeness of generated radiology reports?

3. How effectively can uncertainty estimation mechanisms reflect the reliability of generated radiology reports in clinical decision-support settings?

The main contributions of this work are:

1. a novel hybrid retrieval-augmented framework for radiology report generation that incorporates both visual and contextual knowledge

2. an attention-based multimodal fusion strategy that improves report coherence and completeness

3. an integrated uncertainty estimation and validation pipeline for assessing the reliability of generated reports. Together, these contributions provide a more robust and clinically applicable solution for automated radiology reporting.

We expect that integrating retrieval mechanisms and multimodal fusion will significantly improve the quality of generated reports. Specifically, the proposed approach is anticipated to produce more accurate, coherent, and clinically relevant reports by leveraging contextual knowledge from similar cases. Additionally, the uncertainty estimation module is expected to provide meaningful confidence scores that correlate with the correctness of the generated outputs. Overall, the system should outperform traditional approaches in both quantitative metrics (e.g., BLEU, ROUGE, BERTScore) and qualitative clinical assessment.

2 Related Work

XrayGPT (Thawakar et al., 2024) is a conversational vision-language model (VLM) specifically designed to analyze chest radiographs and answer open-ended medical questions with follow-up capabilities. The model was developed using large-scale datasets, including MIMIC-CXR (377k images), the OpenI dataset, and ChatDoctor for fine-tuning specialized medical dialog. The implementation involves a two-stage training process, which involves aligning a medical vision encoder with an LLM using 217k interactive summaries, followed by fine-tuning on high-quality, curated data. XrayGPT outperformed the Mini-GPT-4 baseline with a ROUGE-1 score of 0.3213 and received a scientific accuracy rating of 72% from certified medical doctors. However, the model's primary weaknesses include decreased accuracy on side-view X-rays, a lack of support for non-chest imaging (like MRI or CT), and occasional mischaracterization of subtle abnormalities.

RADAR (Hou et al., 2025) is a specialized framework designed to improve the accuracy and informativeness of automated radiology reports by integrating an LLM's internal knowledge with externally retrieved data. The model leverages large-scale datasets, including MIMIC-CXR (over 162k training samples), CHEXPERT, and IU X-RAY, focusing specifically on frontal chest X-ray images. The process first generates "Preliminary Findings" aligned with expert image classifications, then injects "Supplementary Findings" from external databases to enrich the report without creating redundancy. The system is built on the BLIP-3 multimodal backbone, utilizing a SigLIP vision encoder and a Phi-3-mini language model to process visual and textual information efficiently. While RADAR achieves superior clinical accuracy (e.g., 0.346 RG-F_1), it is currently limited to frontal chest X-rays and has yet to be tested on other modalities like CT or MRI.

Fact-Aware Multimodal Retrieval Augmentation for Accurate Medical Radiology Report Generation(Sun et al., 2025) this paper introduces FactMM-RAG, a pipeline designed to improve the factual accuracy of AI-generated radiology reports by using a retrieval-augmented generation approach. The model leverages the MIMIC-CXR and IU-Xray datasets, using RadGraph to extract 200,000 factually aligned image-report pairs for precise training. The system consists of a Universal Multimodal Retriever that finds high-quality reference reports and a Generative Foundation Model (like LLaVA-v1.5) that uses those reports as context. FactMM-RAG demonstrated superior factual correctness, achieving an F1-CheXbert score of 0.520, which represents a 3.8% improvement over standard baselines (LLaVA-v1.5, Med-Frozen). The framework is currently limited to chest X-rays and a single backbone architecture, leaving its effectiveness in other modalities like CT or MRI for future research.

3 Methodology

This section describes the overall methodology of the proposed system for automated radiology report generation. The pipeline follows a multimodal, retrieval-augmented framework that integrates image features, clinical knowledge, and retrieved case information. An overview of the system is illustrated in Figure 1, which presents the end-to-end flow from input data to report generation and evaluation.

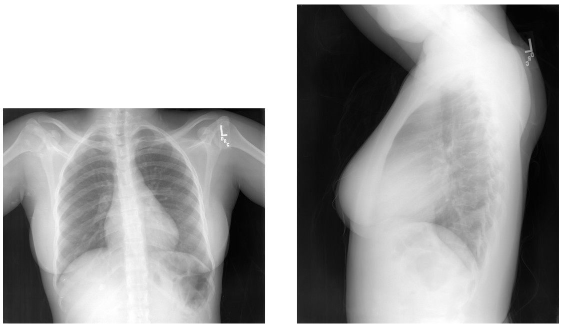
3.1 Dataset

We use the **Indiana University Chest X-ray dataset** [2] for this study. This dataset consists of chest X-ray images and corresponding radiology reports for **3,851 patients**. For each patient, two views are typically available: a *frontal view* and a *lateral view*, which provide complementary anatomical information for clinical interpretation. In addition to images, the dataset includes a CSV file (`indiana_reports.csv`) containing structured and unstructured clinical information. The main columns used in this work are:

- **Problems:** A set of clinically relevant findings identified from the X-ray, representing medical conditions or abnormalities (e.g., cardiomegaly, effusion). These are derived from expert-written radiology reports.
- **Image:** Metadata describing the type or view of the X-ray (e.g., PA, lateral).
- **Indication:** The clinical reason for performing the X-ray (e.g., symptoms or suspected condition).
- **Comparison:** Information about prior studies used for comparison, if available.
- **Findings:** A detailed descriptive section of the radiology report explaining observations in the X-ray.
- **Impression:** A concise summary of the most important clinical conclusions derived from the findings.

For our methodology, the inputs consist of **two X-ray images (frontal and lateral views)** along with the corresponding **Problems** (clinical labels), which provide structured medical knowledge. The target output is the **Findings** section, which serves as the ground truth radiology report to be generated by the model.

[2] https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university



(a) Frontal View (b) Lateral View

Figure 2: Example chest X-ray images from the IU dataset showing (a) frontal and (b) lateral views for a single patient.

3.2 Data Preprocessing

Data preprocessing is a crucial step to ensure consistency and effective utilization of multimodal inputs. The preprocessing pipeline operates on image data, textual data, and structured clinical labels, tailored to the specific components of the proposed hybrid RAG framework.

First, chest X-ray images are organized per patient, ensuring that both **frontal** and **lateral** views are available. These images are resized and normalized according to the requirements of the **BioViL-T** image encoder. The processed images are then used to extract domain-specific visual embeddings for both feature learning and image-based retrieval.

Second, the textual components are processed for two distinct purposes: **report generation** and **text-based retrieval**. For report generation, the **Findings** section (target output) is preprocessed to be compatible with the **R2Gen** model. The text is cleaned by converting to lowercase, removing unnecessary special characters, and normalizing whitespace while preserving clinically relevant terms. A **word-level tokenizer** is used to construct a vocabulary from the training dataset, where each word is mapped to a unique index. Special tokens such as <start>, <end>, and <pad> are added, and each report is converted into a sequence of token IDs. These sequences are then padded or truncated to a fixed maximum length for training.

For text-based retrieval, radiology reports are additionally encoded using **BiomedBERT (PubMedBERT-style) sentence embeddings**. The text is minimally preprocessed and passed through the pre-trained encoder to obtain dense semantic representations, enabling retrieval of clinically similar reports based on meaning rather than exact wording.

Third, the **Problems** column is processed to extract structured clinical knowledge. Each entry, which may contain multiple conditions separated by delimiters, is split into individual labels. These labels are converted into a **multi-hot encoded vector**, allowing the model to represent multiple co-existing conditions. These structured representations are used both as inputs to the model and as an additional signal in the hybrid retrieval process.

During pre-processing, we exclude cases where the **Problems** field is missing or empty, as these labels are essential for retrieval and multimodal fusion. Additionally, only samples with both frontal and lateral images and valid **Findings** are retained. After filtering, the final dataset contains **3,337 patient records**, each with valid frontal and lateral images, associated problem labels, and corresponding findings.

This preprocessing pipeline ensures alignment between all modalities and supports the integration of image features, textual semantics, and structured clinical knowledge within the proposed framework.

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3.3 Models

To evaluate the effectiveness of the proposed method, we compare our approach against several established baseline models for chest X-ray report generation. First, we use **R2Gen** (Chen et al., 2020) as the primary baseline, as it is a widely adopted standard model that generates reports directly from image features. It provides a strong reference point for image-to-text generation. We further include **R2GenCMN** (Chen et al., 2021), which extends R2Gen with cross-modal memory, enabling more informed report generation and serving as a stronger baseline. Additionally, we consider **PPKED** (Liu et al., 2021), a knowledge-enhanced model that incorporates external medical knowledge, making it suitable for evaluating the impact of knowledge integration. Finally, we include **METransformer** (Wang et al., 2023), a modern transformer-based architecture, to ensure comparison with recent and competitive approaches.

Our proposed approach builds upon **R2Gen as the report generation backbone** and extends it with a **hybrid Retrieval-Augmented Generation (RAG) framework**. For image representation and retrieval, we employ **BioViL-T** (Bannur et al., 2023) as the image encoder to extract domain-specific visual embeddings from chest X-rays. For textual retrieval, we utilize **BiomedBERT (PubMedBERT-style) sentence embeddings** (Gu et al., 2021)to capture semantic similarity between radiology reports.

The system performs **multi-signal hybrid retrieval** by combining image-based similarity, text-based similarity, and structured clinical labels (Problems). Retrieved items are treated as **case-level units** (image, report, and labels), ensuring clinical consistency. These multimodal features are integrated through an **attention-based fusion mechanism**, which dynamically learns the importance of different modalities and retrieved cases. The fused representation is then passed to the R2Gen generator to produce the final radiology report. This design enables the model to effectively leverage both learned representations and external contextual knowledge, improving clinical accuracy and reducing hallucinations.

3.4 Proposed Method Overview

Figure 1 illustrates the overall architecture of the proposed system. The framework follows a retrieval-augmented multimodal pipeline that integrates visual features, textual knowledge, and structured clinical information to generate accurate radiology reports.

Input Representation The system takes as input **chest X-ray images** (frontal and lateral views) along with corresponding **clinical problem labels**. These problem labels represent structured medical knowledge and are encoded as multi-label vectors.

Vision Encoder The input X-ray images are passed through a vision encoder, Convolurional Neural Network (CNN) to extract high-level visual feature representations. These features capture anatomical and pathological patterns present in the images.

Hybrid Retrieval (Core Novelty) Unlike traditional approaches that rely on a single retrieval signal, our method employs a **hybrid multi-signal retrieval mechanism**. Given the encoded image features and clinical labels, the system retrieves top-K similar cases from the dataset using:

- **Visual similarity** (image embeddings),
- **Semantic similarity** (retrieved reports),
- **Structured clinical similarity** (problem labels).

Case-Level Retrieval Each retrieved item is treated as a complete **case**, consisting of *image features, corresponding report, and associated problem labels*. This ensures clinical consistency by preserving relationships between modalities, rather than mixing unrelated information from different samples.

Structured Clinical Knowledge Integration The **Problems** column is encoded into multi-hot vectors and incorporated into both retrieval and generation stages. This explicit integration of structured medical knowledge improves the model's ability to capture clinically relevant patterns.

Evidence Fusion (Attention-Based) The retrieved cases, along with the original input features, are combined using an **attention-based multimodal fusion mechanism**. Instead of simple concatenation, the model learns:

- which retrieved cases are most relevant,
- which modality (image, text, or labels) should be emphasized.

This dynamic weighting enables more coherent and context-aware representations.

Multimodal Report Generation The fused representation is passed to the **R2Gen-based language model**, which generates the final radiology report (Findings). The use of retrieval-augmented context ensures that generation is grounded in similar prior cases, improving factual consistency and reducing hallucinations.

Uncertainty Estimation In addition to generating reports, the system computes a **confidence score** for each output. This provides an estimate of prediction reliability, which is critical for real-world clinical applications.

Faithfulness and Hallucination Reduction By grounding the generation process in retrieved evidence, the proposed method improves **faithfulness** and reduces hallucinations. The generated reports are more consistent with both the input images and clinically similar historical cases.

3.5 Training Strategy

The proposed model is trained in a supervised manner using paired chest X-ray images and corresponding radiology reports. The **Findings** section serves as the ground truth target for report generation.

Dataset Split The preprocessed dataset of 3,337 samples is divided into three subsets:

- **Training set (70%)**: Used to learn model parameters.
- **Validation set (10%)**: Used for hyperparameter tuning and model selection.
- **Test set (20%)**: Used for final performance evaluation.

The split is performed at the patient level to avoid data leakage between sets.

Training Procedure During training, each sample consists of frontal and lateral X-ray images, structured problem labels, and the corresponding Findings report. The images are encoded using **BioViL-T**, while the hybrid retrieval module retrieves top-K similar cases based on image, text, and label similarity. The retrieved cases, along with the original inputs, are fused using an attention-based mechanism and passed to the **R2Gen** generator.

The model is trained to generate the target report in a sequence-to-sequence manner. At each time step, the generator predicts the next word conditioned on previous ground truth tokens (teacher forcing).

Loss Function We use the standard **cross-entropy loss** for sequence generation, which measures the difference between the generated token probabilities and the ground truth tokens in the Findings reports.

Optimization The model is optimized using the **Adam optimizer**, which adapts learning rates for efficient convergence. Early stopping is applied based on validation performance to prevent overfitting.

Validation Strategy A validation set is essential to monitor model performance during training. Metrics such as BLEU and validation loss are computed after each epoch. The model with the best validation performance is selected for final evaluation.

Regularization and Training Techniques To improve generalization and stability, the following techniques are applied:

- **Teacher Forcing:** Helps stabilize sequence generation during training.
- **Early Stopping:** Stops training when validation performance stops improving.
- **Dropout:** Applied within the model to reduce overfitting.
- **Gradient Clipping:** Prevents exploding gradients in sequence models.

This training strategy ensures effective learning of multimodal representations while maintaining generalization and robustness across unseen data.

4 Experiments

This section describes the experimental setup used to evaluate the proposed framework, including evaluation metrics, baseline comparisons, and ablation studies.

4.1 Evaluation Metrics

To evaluate the quality of generated radiology reports, we use **BLEU-2 (BL-2)**, **BLEU-3 (BL-3)**, and **BLEU-4 (BL-4)** to measure n-gram overlap between generated and ground truth reports. Additionally, **ROUGE-L** is used to assess recall-based similarity and capture the longest common subsequence between texts. These metrics provide an effective evaluation of the linguistic quality and consistency of the generated reports.

4.2 Baseline Comparison

We compare the proposed model against several established baselines for chest X-ray report generation discussed in Section 3, including **R2Gen**, **R2GenCMN**, **PPKED**, and **METransformer**. These models represent standard, memory-based, knowledge-enhanced, and modern transformer-based approaches, respectively.

Model	BL-2	BL-3	BL-4	ROUGE-L
R2Gen	–	–	–	–
R2GenCMN	–	–	–	–
PPKED	–	–	–	–
METransformer	–	–	–	–
Ours **(R2Gen + Hybrid RAG)**	–	–	–	–

Table 1: Comparison of the proposed model with baseline methods using selected evaluation metrics.

All models are evaluated under the same experimental setup. We report performance using selected evaluation metrics, including intermediate benchmark levels (BL-2, BL-3, BL-4) and **ROUGE-L**, which measures the overlap between generated and ground truth reports.

4.3 Ablation Study

To evaluate the effectiveness of key components in the proposed framework, we conduct an ablation study using a simplified set of model variants.

- **(A0): R2Gen (Baseline):** Uses only image features for report generation.
- **(A1): A0 + Problems:** Incorporates structured clinical labels (Problems) along with image features.
- **(A2): A0 + Retrieval:** Adds retrieval of similar cases based on image features and textual similarity.
- **(A3): R2Gen + Hybrid RAG:** Includes hybrid retrieval, structured labels, and attention-based fusion.

Model	BL-2	BL-3	BL-4	ROUGE-L
(A0) R2Gen	–	–	–	–
(A1) + Problems	–	–	–	–
(A2) + Retrieval	–	–	–	–
(A3) Full Model	–	–	–	–

Table 2: Ablation study evaluating the contribution of structured clinical labels and retrieval mechanisms.

All variants are evaluated using BLEU-2, BLEU-3, BLEU-4, and ROUGE-L. This study highlights the incremental improvements achieved by incorporating structured clinical knowledge and retrieval-based augmentation.

5 Conclusion

In this proposal, we present a hybrid retrieval-augmented framework for automated radiology report generation that integrates visual features, structured clinical knowledge, and retrieved case-based evidence. The proposed approach builds upon R2Gen as the report generation backbone and enhances it with multi-signal retrieval using BioViL-T for image encoding and BiomedBERT for text-based similarity.

The key idea of this work is to leverage hybrid retrieval, case-level consistency, and attention-based multimodal fusion to improve the quality and clinical relevance of generated reports. Additionally, the inclusion of structured problem labels and uncertainty estimation aims to enhance both interpretability and reliability in clinical decision-support settings.

We expect that the proposed method will outperform traditional image-to-text approaches by generating more coherent, context-aware, and clinically grounded reports. Future work will involve implementing the full pipeline, conducting extensive experiments, and evaluating the model against established baselines using standard metrics.

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